

Physics-Informed Neural Networks for Transient Thermal Analysis and Insulation Life Assessment in Power Transformers

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Abstract—Transformers are increasingly being subjected to dynamic, non-linear loading profiles driven by the stochastic nature of renewable energy integration and the high power demands of electric vehicle charging infrastructure. The primary limiting factor of transformer lifespan is the thermal degradation of its solid insulation system, typically oil impregnated cellulose. Precise real time Hot-Spot Temperature (HST) estimation is therefore critical for transformer reliability, for which traditional numerical methods are computationally prohibitive while standard data-driven models lack physical consistency. This paper proposes a novel hybrid Physics-Informed Neural Network (PINN) framework that integrates the governing transient heat diffusion partial differential equations (PDEs) directly into the deep learning architecture. The model is integrated with the recommendations of IEEE standards to provide a continuous real-time estimate of the aging Acceleration Factor (F_{AA}), enabling proactive monitoring of the insulation life of the transformer.

Index Terms—Power Transformers, Physics Informed Neural Networks (PINN), Finite Element Method (FEM), Hot Spot Temperature (HST), Insulation Aging

I. INTRODUCTION

The power transformer is potentially the most capital-intensive and operationally critical asset in electrical power systems. Transformer winding insulation, which is typically composed of cellulose-based kraft paper or pressboard impregnated with mineral oil, provides both electrical isolation and mechanical support to the windings, playing a crucial role in the actual operational life of the transformer. The degradation of this type of insulation is an irreversible chemical reaction, known as depolymerization. As the cellulose molecular chains break down, the mechanical strength of the insulation diminishes [1], [2]. If the Degree of Polymerization (DP) falls below a certain level ($DP < 200$), the insulation becomes brittle. In such cases, the electromechanical stresses generated by a general through-fault or a switching surge can cause the insulation to fail, leading to an inter-turn short circuit and ultimately catastrophic failure.

The insulation degradation in transformers is primarily caused by three factors: thermal stress (pyrolysis), moisture (hydrolysis), and oxygen (oxidation), with thermal stress being the prominent factor under normal operating conditions [3].

Traditional transformer design relies on steady-state assumptions that are increasingly challenged by the changing dynamics of smart grids and modern energy technologies. Distributed Energy Resources (DERs), such as solar photovoltaic and wind energy, enable bidirectional power flow and high ramp rates. The electrification of the transportation sector causes high-magnitude, coincident peak loads due to Electric Vehicle (EV) charging. These transient phenomena can cause localized heating, or “Hot Spots”, which exceed the design capacity of conventional cooling systems, thus accelerating insulation degradation at localized points in the winding. The standard thermal models, such as the top-oil and hot-spot equations mentioned in IEEE C57.91 [4], use simplified lumped-parameter models. While computationally efficient, these models often do not accurately represent the complex spatial and temporal thermal gradients in the winding geometry, especially during rapid transients [5], [6].

Thermal monitoring in modern transformers can be done either through direct measurement using fiber optics or through Finite Element Method (FEM) modelling [7]. While fiber-optic sensors are immune to electromagnetic interference and provide real-time accuracy, their high cost and pre-manufacturing installation requirements render them impractical for traditional transformers. On the other hand, Finite Element Modeling is the standard for accurate 3D thermal modelling, achieved by solving fluid-thermal equations; however, its great computational intensity also causes appreciable latency. This makes Finite Element Modeling unsuitable for decision-making at the millisecond timescale, which is necessary for edge computing and real-time digital twin simulations. Thus, there is still a gap between the accuracy of numerical models and the need for speed in proactive asset management [8]–[11].

To bridge the gap between accurate Finite Element Methods (FEM) and efficient Lumped Parameter models, the present work uses Physics-Informed Neural Networks (PINNs) [12]. By incorporating the governing Partial Differential Equations (PDEs) into the loss function, PINNs overcome the “black-box” constraints that come with traditional deep learning [13]. This allows PINNs to provide three key benefits: ex-

ceptional data efficiency in sparse sensor configurations, strict adherence to thermodynamic laws during extrapolation, and mesh independence for continuous spatio-temporal evaluation. Therefore, PINNs help to resolve discretization errors while maintaining the computational speed required for real-time transformer monitoring and health management.

II. TRANSFORMER MODELING

A. Geometry and Domain Definition

The thermal performance was analyzed by using a 2D cross-section of the transformer winding and the surrounding oil duct, which is common in thermal analysis, as the radial heat flux is normally dominant over the axial heat flux at the mid-height of the transformer winding. In the context of FEM, the term “Face” is a specific label used to identify a 2D bounded region within a geometry. The geometry in the present work is divided into two distinct domains. The first domain represents the mineral oil (Face-1 or F1), which is a 10 cm × 10 cm square. This domain acts as the cooling agent as well as the dielectric insulator. The convection cooling of the winding by oil domain is modelled using an effective thermal conductivity [14]. The second domain represents the copper winding (Face-2 or F2), which is a 2 cm × 2 cm square and is placed at the center of the oil domain. This domain is the source of volumetric heat generation.

The geometric representation of the transformer consists of a combination of the above-mentioned regions, as shown in Fig. 1. This spatial division allows the PDE solver to independently apply different thermal properties and boundary conditions to the coolant and insulation subsystems.

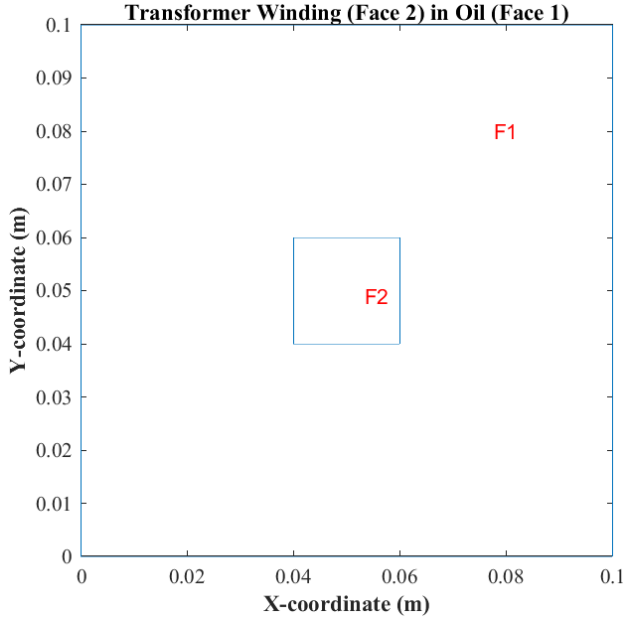


Fig. 1. Two-dimensional geometric representation of the transformer thermal domain, the surrounding mineral oil (F1) and the heat-generating copper winding (F2).

B. Material Properties and Thermodynamics

The thermal response of the system is characterized by the material-dependent thermal conductivity and heat capacity, which control the rate of heat diffusion and the thermal inertia of the oil and copper regions. The thermal properties assigned to the geometric domains are listed in Table I. The properties of mineral oil (F1) and copper winding (F2) are chosen following typical manufacturer specifications and IEEE thermal modelling recommendations [15]. In particular, the selected copper properties are associated with high-conductivity Electrolytic Tough Pitch (ETP) copper, while the oil properties are attributed to inhibited naphthenic mineral oil at a nominal operating temperature of 70°C. These parameters are chosen to ensure the physical correctness of the Heat Diffusion Equation in the PINN formulation.

TABLE I
MATERIAL PROPERTIES FOR GEOMETRIC DOMAINS

Property	Mineral Oil (Face 1)	Copper Winding (Face 2)
Thermal conductivity, k (W/m·K)	0.13	400
Mass density, ρ (kg/m ³)	880	8960
Specific heat, c_p (J/kg·K)	1900	385

The large difference in thermal conductivity between the copper winding and mineral oil causes a strong temperature gradient at the interface between the two materials, which poses a challenge for numerical convergence. Moreover, the high volumetric heat capacity of the oil domain (ρc_p) causes a large thermal inertia, which affects the time-constant of the cooling response of the transformer.

C. Governing Equations

The transient temperature distribution $T(x, y, t)$ is governed by the parabolic partial differential equation for heat diffusion as given here:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q \quad (1)$$

where ρ is the mass density, c_p is the specific heat capacity, and the term $\partial T / \partial t$ represents the rate of energy storage in the material medium. The term $\nabla \cdot (k \nabla T)$ describes the spatial heat diffusion process following Fourier’s Law, which allows for discontinuities in thermal conductivity k at material interfaces. The volumetric internal heat source Q (W/m³) describes the thermal energy produced by the combined Joule and eddy current losses. In the current work, a constant heat source of 100 kW/m³ is applied exclusively to the winding domain, while the oil domain is treated as a passive medium with $Q = 0$.

To ensure a well-posed boundary value problem, the system is modelled as a “cold start” scenario with a uniform initial condition of $T(x, y, 0) = 25^\circ\text{C}$ across both domains. The external boundaries of the oil domain are governed by Dirichlet conditions, where the temperature is held constant at 25°C to

simulate a transformer tank wall acting as an infinite heat sink in thermal equilibrium with the ambient environment.

III. PHYSICS-INFORMED NEURAL NETWORK FRAMEWORK

Standard deep learning techniques approximate a function $f(x)$ by minimizing the error between the network output and labelled training data. However, this approach ignores the physics governing the data. PINNs modify this by defining the residual of the PDE as a regularization term. If the network output $T(x, y, t)$ satisfies the heat equation, the residual will be zero. Any deviation from zero indicates that the laws of physics under consideration are violated, which is penalized proportionately during training [16], [17].

A. Neural Network Architecture

The PINN used in the present work is a fully connected deep network (multilayer perceptron) that aims to estimate the scalar field $T(x, y, t)$. The network architecture includes an input layer with three neurons representing the spatiotemporal coordinates (x, y, t) and an output layer with a single neuron that provides the estimated temperature. In order to model the strongly nonlinear temperature patterns in the transformer domains, three hidden layers with 50 neurons in each are used. The hyperbolic tangent activation function ($\tanh$) is used in all hidden layers to make the network infinitely differentiable. These hyperparameter selections were done based on PINN literature and empirical evaluation. This choice is critical for the PINN framework, as it allows for the accurate computation of the spatial Laplacian ($\nabla^2 T$) required by the heat diffusion equation; unlike the Rectified Linear Unit (ReLU), the $\tanh$ function provides the continuous second-order derivatives necessary for evaluating the physics-based loss function.

B. Hybrid Loss Function Formulation

The training of the Physics-Informed Neural Network (PINN) is formulated as an unconstrained optimization problem to minimize a composite loss function L_{total} :

$$L_{total} = L_{data} + \lambda_{PDE} L_{PDE} \quad (2)$$

where λ_{PDE} is a weighting hyperparameter. The individual loss terms are defined as follows:

1) *Data Loss (L_{data})*: This term enforces the agreement between the network predictions and the available sparse measurement data. In this study, only 10% of the Finite Element Method (FEM) nodal data is utilized to simulate sparse sensor availability. The loss is calculated as the Mean Squared Error (MSE):

$$L_{data} = \frac{1}{N_d} \sum_{i=1}^{N_d} |T_{pred}(x_i, t_i) - T_{meas}(x_i, t_i)|^2 \quad (3)$$

where N_d is the number of data points, T_{pred} is the PINN prediction, and T_{meas} is the measured temperature.

2) *Physics Loss (L_{PDE})*: This term enforces physical consistency by penalizing deviations from the governing physics [18]. The PDE residual f_{res} is evaluated at a set of N_f collocation points, which are distinct from the data measurement points:

$$f_{res} = \frac{\partial T}{\partial t} - \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) - Q_{norm} \quad (4)$$

The physics loss is then defined as the MSE of this residual:

$$L_{PDE} = \frac{1}{N_f} \sum_{j=1}^{N_f} |f_{res}(x_j, t_j)|^2 \quad (5)$$

where α is the thermal diffusivity and Q_{norm} is the normalized heat source term. The weighting hyperparameter was set to $\lambda_{PDE} = 0.1$ following empirical evaluation, which yielded the lowest PDE residual during training convergence.

C. Derivative Computation: Hybrid Approach

An important aspect of the implementation of PINNs is the approach used to evaluate partial derivatives in the loss function [19]. The present work uses a hybrid approach to balance accuracy and stability as follows:

1) *Temporal Derivative ($\frac{\partial T}{\partial t}$)*: It is computed using Automatic Differentiation (AD) via MATLAB's `dlgradient` function. AD applies the calculus chain rule to the computational graph of the neural network, providing precise derivatives without the inherent discretization errors present in numerical methods.

2) *Spatial Laplacian ($\nabla^2 T$)*: It is computed using a second-order central Finite Difference (FD) approximation. For example, the second derivative with respect to x is approximated as follows:

$$\frac{\partial^2 T}{\partial x^2} \approx \frac{T(x+h, y, t) - 2T(x, y, t) + T(x-h, y, t)}{h^2} \quad (6)$$

where $h = 0.001$ is the perturbation step. Although AD is also applicable for spatial derivatives, the FD method can offer better stability for problems involving material discontinuities or "stiff" gradients, as the finite perturbation step h can have a smoothing effect.

IV. SIMULATION SETUP AND DATA GENERATION

A. Finite Element Baseline

To test the PINN, a ground truth dataset was generated using the Finite Element Method (FEM) using the MATLAB PDE Toolbox. The discretization of the domain was performed using a triangular mesh with a maximum element size H_{max} of 0.005 m. This resolution ensures that the thermal gradients at the winding-oil interface are captured accurately. The transient solver was set to operate for 3600 seconds (1 hour) with 30-second time steps. This duration is sufficient to observe the transformer heating from ambient to steady-state thermal equilibrium.

B. Data Pre-processing

The raw inputs span different orders of magnitude: Time (0 – 3600), Space (0 – 0.1), and Temperature (25 – 80). Training on raw data can lead to unstable gradients and poor convergence, as neural networks are sensitive to the scale of the input data. Therefore, Min-Max normalization was applied to scale all inputs and outputs to the range [0, 1]:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (7)$$

C. Subsampling Strategy

A robust AI model for transformers must operate with limited data. To simulate this, the dense FEM dataset (containing thousands of nodal values) was randomly subsampled. Only 10% of the data points were used for training. The rest of the 90% data points were used for validation and testing purposes to ensure that the model does not overfit on the training data.

V. RESULTS AND DISCUSSION

A. Training Dynamics and Convergence

The Adam (Adaptive Moment Estimation) optimizer was used to train the network for 1000 epochs at a learning rate of 0.001, due to its capacity to manage non-stationary objectives and sparse gradients. The PINN achieved a training time of 608.5 seconds. Two separate phases are revealed by the loss curve analysis, as illustrated in Fig. 2. In the Data Fitting Phase (Iterations 0–200), the L_{data} term drops rapidly as the network learns the coarse shape of the thermal distribution from the sparse sensor points. The L_{PDE} term diminishes more slowly during the Physics Refinement Phase (Iterations 200–1000), adjusting the gradients between data points to guarantee that the temperature field curvature satisfies the Laplacian operator of the heat equation. This serves as a useful guide for the optimization process, as demonstrated by the successful convergence of both terms.

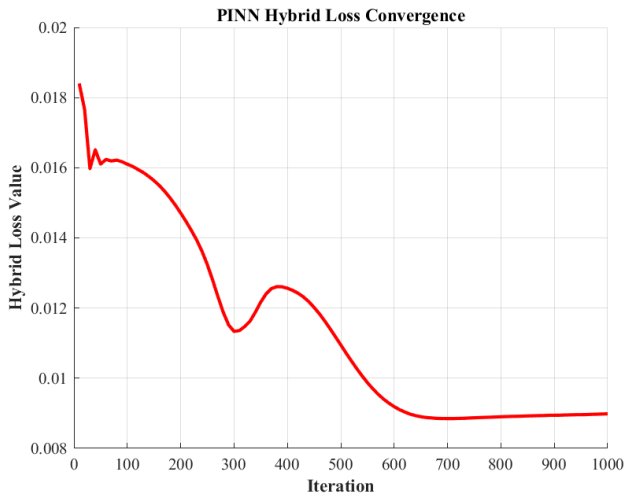


Fig. 2. Convergence profile of the PINN hybrid loss function over 1000 iterations.

B. Thermal Field Reconstruction

The diffusion profile displayed by the FEM simulation is shown in Fig. 3. The temperature decreases radially outward through the oil to the surrounding tank walls, with the copper winding core being the hottest area (Hot-Spot). This FEM baseline is used as the ground truth for testing the capability of the PINN model to reconstruct the thermal gradient between the copper winding and the insulating oil. The PINN model successfully captures the sharp change in the temperature gradient at the winding-oil interface, as shown in Fig. 4. Standard neural networks often fail to properly capture such discontinuities. The Physics Loss component forces the network to consider the change in thermal conductivity k , resulting in a sharp and physically accurate interface. Fig. 3 (FEM baseline) and Fig. 4 (PINN Prediction) provide a qualitative comparison of the thermal fields at steady state ($t = 3600$ s).

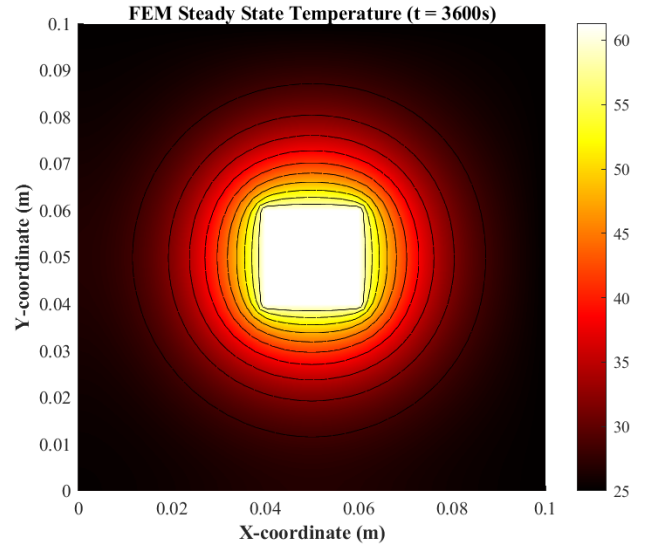


Fig. 3. Steady-state temperature distribution ($t = 3600$ s) obtained via Finite Element Method (FEM).

C. Quantitative Error Metrics

To quantify the performance, the Root Mean Square Error (RMSE) was calculated across the entire spatial-temporal domain, which gave a value of 4.1430°C . Given that the typical allowable temperature rise for a transformer is 65°C over ambient (reaching $\sim 90 - 100^{\circ}\text{C}$ absolute), an error of $\sim 4^{\circ}\text{C}$ represents approximately 4–5% relative error. This falls well within the acceptable margins for industrial thermal protection relays, which often have calibration uncertainties of 2–3°C [20].

The Absolute Error Map shown in Fig. 5 provides critical insight into the limitations of the model. The large oil domain shows near-zero error, as the physics of simple diffusion in a homogeneous medium is easier for the network to learn. The highest errors are localized strictly at the boundaries of the copper winding. This “boundary layer” error is a known

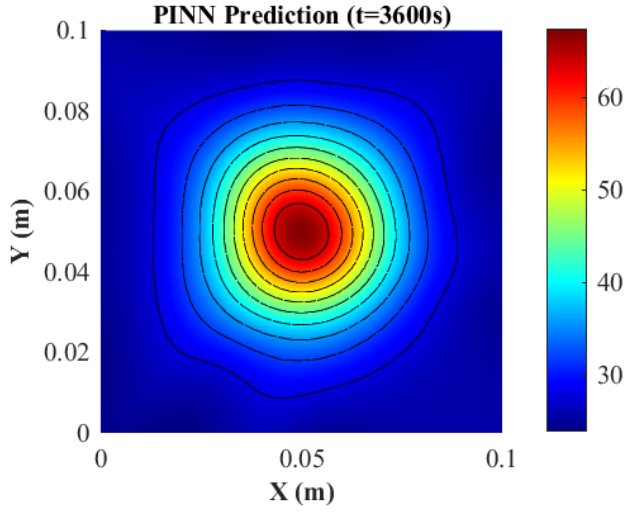


Fig. 4. Thermal field reconstruction by the PINN model at $t = 3600$ s.

phenomenon in numerical methods, caused by the Gibbs phenomenon where approximating a discontinuous function (the jump in material properties) with smooth basis functions (Neural Network activations) leads to oscillations [21].

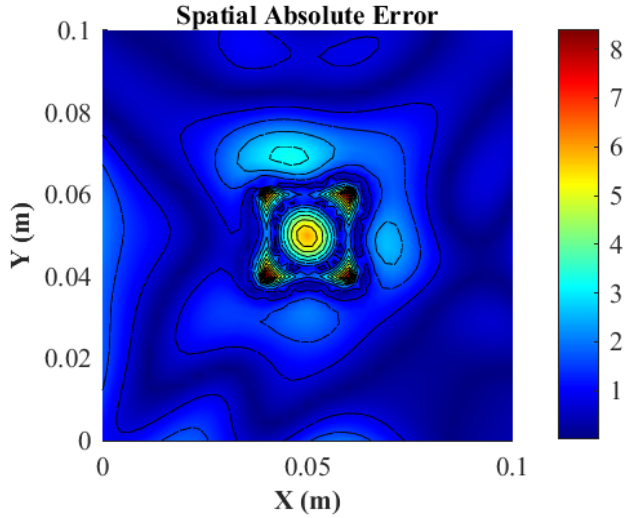


Fig. 5. Absolute error map between PINN prediction and FEM ground truth at $t = 3600$ s.

D. Temporal Evolution and Transient Response

The most critical aspect for dynamic rating is the ability to track temperature changes over time. The comparative plots of HST Evolution show the temporal trajectory of the hot-spot, as shown in Fig. 6. The results show high temporal fidelity in tracking the transformer's heating curve from initial state to steady-state equilibrium. The PINN accurately tracks the exponential rise characteristic of first-order thermal systems. The model correctly identifies the thermal time constant τ of the system, ensuring that the predicted heating curve aligns with the physics of the thermal mass (ρc_p).

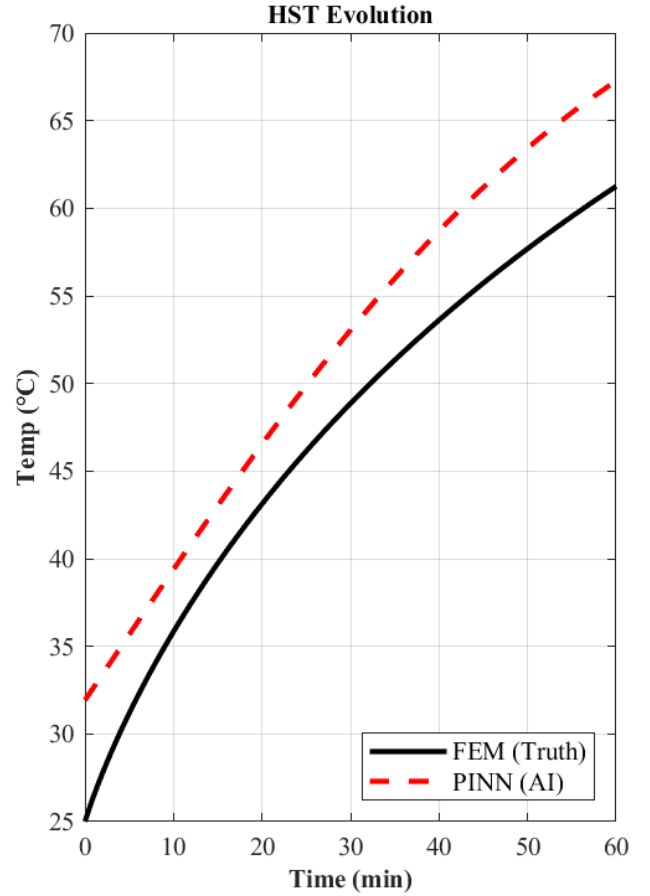


Fig. 6. Comparison of Hot-Spot Temperature (HST) transient evolution between the FEM baseline (truth) and PINN prediction.

VI. TRANSFORMER INSULATION LIFE ASSESSMENT

A. The Arrhenius Degradation Model

The practical value of HST prediction lies in its translation to assessment of insulation life. The degradation of cellulose is modelled as a chemical reaction rate problem using the Arrhenius equation. The IEEE C57.91 standard defines the Aging Acceleration Factor (F_{AA}). F_{AA} is a unitless multiplier that normalizes the aging rate to a reference temperature of 110°C [22].

- If $HST = 110^\circ\text{C}$, $F_{AA} = 1.0$ (1 hour of operation = 1 hour of life lost).
- If $HST > 110^\circ\text{C}$, $F_{AA} > 1.0$ (Accelerated aging).
- If $HST < 110^\circ\text{C}$, $F_{AA} < 1.0$ (Preserved life).

The equation used is:

$$F_{AA} = \exp \left(\frac{15000}{383.15} - \frac{15000}{\theta_{HST} + 273.15} \right) \quad (8)$$

where θ_{HST} is the winding hot-spot temperature in Celsius. The constant 15000 represents the activation energy of the cellulose degradation reaction divided by the gas constant.

Using the HST profiles generated by the PINN, the F_{AA} was calculated for the simulation period. The results are visualized in Fig. 7. The PINN-derived aging curve (red dashed line) tracks the FEM-derived curve (black solid line) with high fidelity. This result validates the use of PINNs for Prognostic Health Management (PHM). The close agreement illustrates the model's reliability for insulation health monitoring and remaining useful life (RUL) estimation. A utility operator could integrate this PINN model into a SCADA system to receive real-time feedback on how a specific overload event (e.g., an EV charging peak) impacts the remaining life of the transformer asset. Unlike FEM, which would take significant time to calculate this impact, the PINN provides the F_{AA} value instantly.

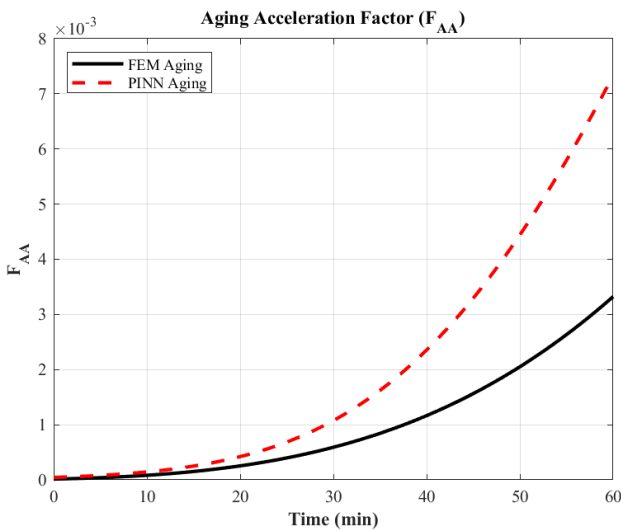


Fig. 7. Comparison of the Aging Acceleration Factor (F_{AA}) derived from FEM and PINN thermal profiles.

VII. CONCLUSION

A comprehensive framework for applying Physics-Informed Neural Networks (PINNs) to the domain of power transformer thermal monitoring was studied in the present work. By bridging the gap between high-fidelity numerical simulation (FEM) and agile data-driven learning, PINNs offer a “best-of-both-worlds” solution. The PINN successfully reconstructed the full thermal field using only 10% of the training data, proving its viability in sensor-sparse environments. The obtained RMSE of 4.1430°C indicates sufficient precision for thermal protection and dynamic rating applications. The inclusion of the physics loss component ensured that the model considered the thermodynamics of heat transfer, particularly at the challenging material interface between copper and oil. The immediate practical utility of the proposed method was demonstrated by the successful computation of the Aging Acceleration Factor (F_{AA}).

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